



Lab: DNS (3%)

Overview:

This lab focuses on DNS servers. You'll install, configure, and manage DNS servers on all VMs in your daisy-chain network.

Objectives and Tools:

- Design/create/deploy a three (3) VM daisy-chain topology that includes Windows and Linux VMs.
- Deploy DNS servers on all VMs and assign FQDN to each VM in the topology.
- Understand how to install, configure, and manage DNS servers (Root, Primary, Secondary, Cache) and the various DNS record types.
- Describe the different types of DNS: Root, Authoritative, Primary, Secondary, and Cache DNS servers.
- Understand forward/reverse lookup zones and forward/reverse lookup queries.
 - Note: For your DNS information to work correctly, each machine must have a stable ip address. You may either revert to static ip addressing in each machine's netplan files, or provide a fixed address through DHCP.
- Demonstrate zone transfer, and verification of DNS deployment based on network traffic analysis.
- Manage and control DNS traffic

Lab Activities

Primary Goals:

Design a three (3) VM daisy-chain topology that includes Windows and Linux computers:

- The Gateway will act as a caching forwarder for the machines in your network, the end node will act as a primary DNS server.
 - The primary DNS server (and the secondary, if configured – see secondary goals) forward all queries they can't answer to the caching server.
- All VMs have FQDNs: <role>.<Z-value>network.net e.g. gateway.99network.net
- All VMs are authorized to send DNS traffic to local DNS servers only. See notes for Secondary Goals.
- Only authorized network traffic is allowed.
- All computers have Internet access and can ping using IP and FQDN.
- All computers can access each other using IP and FQDN.

Secondary Goals:

- Configure a Secondary DNS server on the Windows VM.

Note: The following secondary goals apply to your Linux VMs only.

- Review and apply the following items from CIS ISC BIND DNS Server 9.11 Benchmark (v1.0.0) chapter 2:
 - 2.4-2.8

- Review and apply the following items from CIS ISC BIND DNS Server 9.11 Benchmark (v1.0.0) chapter 3:
 - o 3.1-3.4

Tools:

- dig, nslookup
- Resolve-DnsName, Get-DnsClient, Set-DnsClientServe, Clear-DnsClientCache

Lab Formatting & Submission

Evaluation of this lab will be done through command output.

On the due date you will be provided with one or more scripts and will upload the output from each to blackboard.

Primary tasks to be successfully demonstrated:

- *The end node has a running DNS service that acts as a primary server for forward and reverse lookups of your assigned network.*
- *The gateway machine will act as a caching forwarder for your other DNS servers. Any queries that the primary or secondary servers are not authoritative for must be sent through the caching DNS service running on the gateway.*
- *All Machines are able to reach each other and the internet using ip addresses and FQDNs*
- *Firewalls are still in place preventing unwanted traffic from being allowed to reach any machine*

Secondary tasks to be successfully demonstrated:

- *The Windows node is acting as a secondary DNS server for forward and reverse lookups of your assigned network.*
- *The Linux DNS servers have the identified CIS Benchmark items applied.*